



BAULKHAM HILLS HIGH SCHOOL

2009

**YEAR 12 HALF YEARLY
EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

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Attempt Questions 1 – 7

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Marks

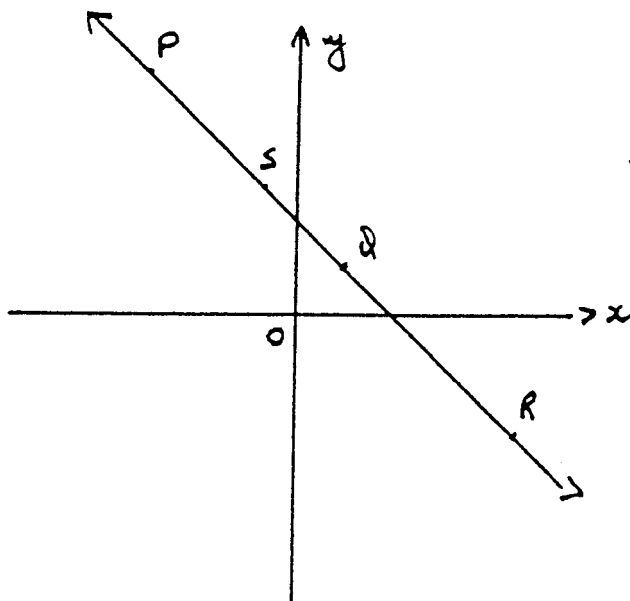
Question 1 (12 marks) Use a separate piece of paper

a) Find the acute angle between the lines $2x + y = 15$ and $3x - y = 7$

2

b)

3



P is the point $(-3, 5)$.

$R(5, -3)$ divides PQ externally in the ratio $2:1$. Find the coordinates of S which divides PQ internally in the ratio $2:1$.

c) Find the Cartesian equation of the curve whose parametric coordinates are

2

$$x = \cos 2t$$

$$y = \cos t$$

d) Evaluate $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{3x}$

2

e) $x^2 - x - 2$ is a factor of $x^4 + 3x^3 + ax^2 - 2x - b$. Find the values of a and b .

3

Question 2 (12 marks) Use a *separate* piece of paper**Marks**

a) The equation $x^3 - 3x^2 + 4x + 2 = 0$ has roots α, β and γ . Find the values of;

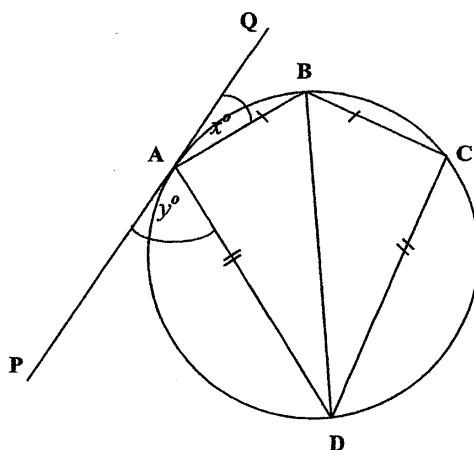
(i) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ 2

(ii) $\alpha^2 + \beta^2 + \gamma^2$ 2

b) (i) On the same diagram sketch $y = x$ and $y = |x - 2|$ 2

(ii) Hence or otherwise solve $\frac{|x-2|}{x} \geq 1$ 2

c) The line PQ is a tangent to the circle at the point A .
 $AB = BC$ and $AD = DC$. $\angle QAB = x^\circ$ and $\angle PAD = y^\circ$



(i) Deduce that $x + y = 90^\circ$ 3

(ii) Explain why BD is a diameter of the circle. 1

Question 3 (12 marks) Use a *separate* piece of paper

a) Find $\int \cos^2 2x dx$ 2

b) If $\tan x = 2$, $0^\circ < x < 90^\circ$ and $\tan y = \frac{-1}{3}$, $90^\circ < y < 180^\circ$, find the exact value of $x + y$ 3

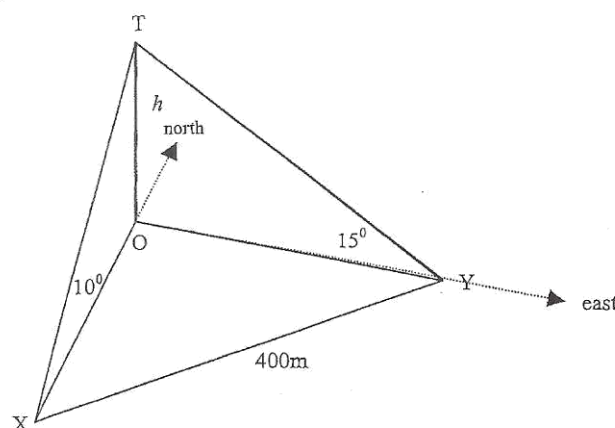
c) Use $x = 0.5$ to find an approximation for the root of $\cos x = x$ using one application of Newton's Method, correct to 2 decimal places. 3

d) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $A \sin(x + \alpha)$ 2

(ii) Hence, or otherwise, solve $\sin x + \sqrt{3} \cos x = 1$ for $0 \leq x \leq 2\pi$ 2

Question 4 (12 marks) Use a *separate* piece of paper**Marks**

- a) Show that $\frac{1 + \cos 2\theta}{\sin 2\theta} \equiv \cot \theta$ 2
- b) A point moves along the curve $y = \frac{1}{x}$ such that the x value is changing at a rate of 2 units per second. 3
At what rate is the y value decreasing when $x = 5$?
- c) A surveyor at X observes a tower due north. The angle of elevation to the top of the tower is 10° . He then walks 400 metres to a position Y which is due east of the tower. The angle of elevation from Y to the top of the tower is 15° .



- (i) Write equations for OX and OY in terms of h . 2
- (ii) Calculate h to the nearest metre. 3
- (iii) Find the bearing of X from Y , to the nearest degree. 2

Question 5 (12 marks) Use a *separate* piece of paper

- a) Solve the equation $3x^3 - 17x^2 - 8x + 12 = 0$ given that the product of two of the roots is 4. 3
- b) (i) Show that $\frac{d}{dx}(\tan^3 x) = 3(\sec^4 x - \sec^2 x)$ 2
- (ii) Hence find $\int \sec^4 x \, dx$ 2
- c) At time t minutes, the temperature T° of a body in a room of constant temperature 20° is decreasing according to the equation $\frac{dT}{dt} = -k(T - 20)$ for some constant $k > 0$
- (i) Verify that $T = 20 + Ae^{-kt}$, where A is constant, is a solution of the equation. 2
- (ii) The initial temperature of the body is 90° and it falls to 70° after 10 minutes. Find the temperature of the body after a further 5 minutes, correct to the nearest degree. 3

Question 6 (12 marks) Use a *separate* piece of paper

- a) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$. The focus is $S(0, a)$.
 PN is perpendicular to the tangent at P . SN is parallel to the tangent at P .

(i) Show that the equation of PN is $x + py = ap^3 + 2ap$ 2

(ii) Show that the coordinates of N are $(ap, ap^2 + a)$ 3

(iii) Find the Cartesian equation of the locus of N . 1

- b) Mayank borrows \$30000 at 9% per annum reducible interest calculated monthly.
 The loan is to be repaid in 60 equal monthly instalments of \$M.

(i) Show that the amount A_n owing after n monthly repayments have been made is 2

$$A_n = 30000(1.0075)^n - M \left(\frac{1.0075^n - 1}{0.0075} \right)$$

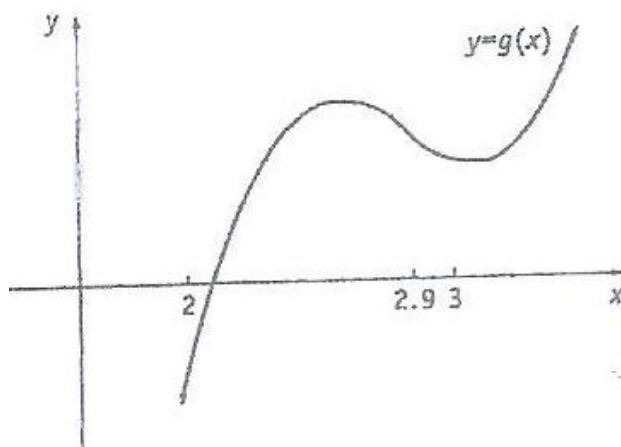
(ii) Show that the monthly repayments should be \$622.75 1

(iii) With the 12th repayment, Mayank pays an additional \$6000, so this repayment is \$6622.75. 3

How many more repayments will be needed?

Question 7 (12 marks) Use a *separate* piece of paper

- a) 2



Consider the above graph of $y = g(x)$. There is a root between $x = 2$ and $x = 3$.
 Explain why $x = 2.9$ is not a suitable first approximation for Newton's method to find this root.

Question 7 continues on page 6

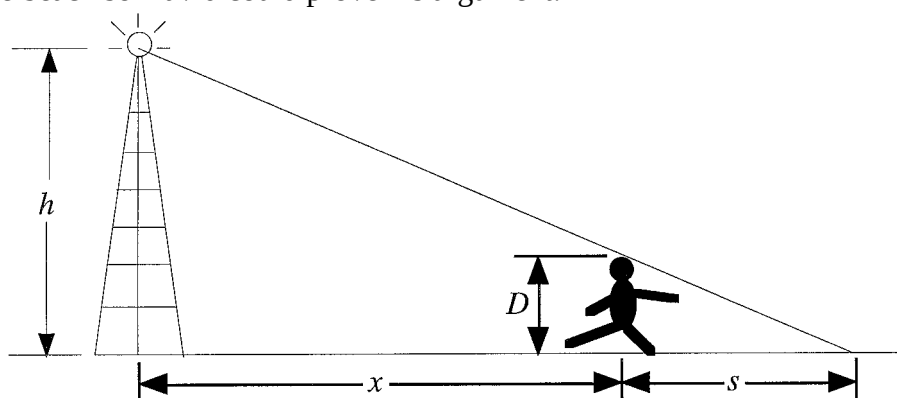
Question 7 (continued)

- b) (i) Use mathematical induction to prove for positive integral values of n ; 3

$$\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \dots + \frac{1}{n(n+2)} = \frac{n(3n+5)}{4(n+1)(n+2)}$$

- (ii) Hence, or otherwise, find $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k(k+2)}$ 1

- c) Two lifeguards, David and Pamela, were having an argument on the beach. David bet Pamela \$50 that he could outrun his own shadow. That night they return to the beach so David could prove his argument.



In the diagram, h represents the height of the lighthouse, D represents David's height, x is David's displacement from the lighthouse and s is the length of David's shadow.

David runs with a velocity of v metres per second.

- (i) Prove that $x = s \left(\frac{h}{D} - 1 \right)$. 2
- (ii) Hence, or otherwise, show that $\frac{ds}{dt} = \frac{Dv}{h-D}$ 2
- (iii) Assuming the lighthouse is 5 metres tall and David is 1.2 metres tall, 1
show that $\frac{ds}{dt} < v$
- (iv) David believed that he had proven his point and stated; "If the velocity of the shadow is always less than my velocity, then eventually I must overtake it." 1
However Pamela saw a mistake in David's argument and refused to pay him \$50. What was David's mistake?

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log x, \quad x > 0$

2009 Extension 1 Half Vrlty Solutions

Question 1 (12)

a) $2x + y = 15 \Rightarrow m_1 = -2$
 $3x - y = 7 \Rightarrow m_2 = 3$

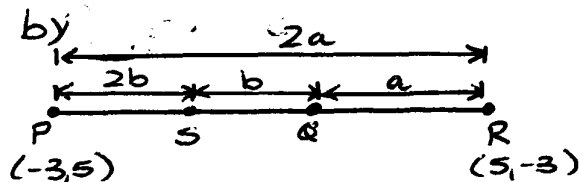
$$\tan \alpha = \left| \frac{-2 - 3}{1 + (-2)(3)} \right|$$

$$= \left| \frac{-5}{-5} \right|$$

$$= 1$$

$$\alpha = 45^\circ$$

(2)



$a = 3b$
 $\therefore S$ divides PR internally
 in the ratio
 $2:4 = 1:2$

$P(-3, 5) \quad R(5, -3)$

$$S = \left(\frac{-6+5}{3}, \frac{10-3}{3} \right)$$

$$= \left(-\frac{1}{3}, \frac{7}{3} \right)$$

(3)

c) $x = \cos 2t \quad y = \cos t$
 $x = 2\cos^2 t - 1$

$$x = 2y^2 - 1$$

(2)

d) $\lim_{x \rightarrow 0} \frac{\sin(\frac{x}{3})}{3x}$
 $= \lim_{x \rightarrow 0} \frac{1}{9} \times \frac{\sin(\frac{x}{3})}{\frac{x}{3}}$
 $= \frac{1}{9}$

(2)

e) $x^2 - x - 2 = (x-2)(x+1)$

$\therefore P(2) = 0 \quad P(-1) = 0$
 $16 + 24 + 4a - 4 - b = 0 \quad 1 - 3 + a + 2 - b = 0$
 $4a - b = -36 \quad a - b = 0$
 $3a = -36 \quad a = b$
 $a = -12$

$\therefore a = -12, b = -12$

(3)

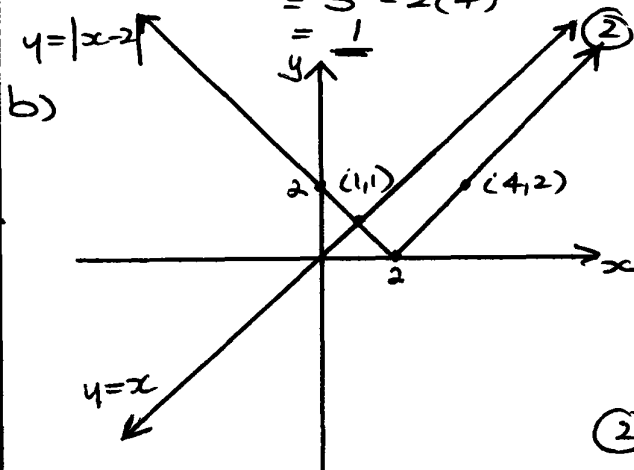
Question 2 (12)

a) $x^3 - 3x^2 + 4x + 2 = 0$

(i) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$
 $= \frac{\gamma + \beta + \alpha}{\alpha\beta\gamma}$
 $= \frac{3}{-2}$

(2)

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 3^2 - 2(4)$
 $= 1$



(2)

(iii) $\frac{|x-2|}{x} \geq 1$

If $x > 0$
 $|x-2| \geq x$
 $0 < x \leq 1$

If $x < 0$
 $|x-2| \leq x$
 no solutions

$\therefore 0 < x \leq 1$

(2)

c) $AB = CB$ (given)
 $AD = CD$ (given)

BD is common

$\therefore \triangle ABD \equiv \triangle CBD$ (SSS)

$\angle BAD = \angle BCD$ (matching $\angle$ s in $\equiv$ Δ s)

$\angle BAD + \angle BCD = 180^\circ$

(opposite $\angle$ s in cyclic quadrilateral)

$\therefore \angle BAD = \angle BCD = 90^\circ$

$x + y + 90^\circ = 180^\circ$ (straight $\angle$ QAP)

$x + y = 90^\circ$

(3)

(ii) $\angle BAD = 90^\circ$

$\therefore BD$ is diameter

($\angle$ in semicircle $= 90^\circ$)

(1)

Question 3 (12)

$$\begin{aligned} \text{a) } \int \cos^2 2x \, dx &= \frac{1}{2} \int (1 + \cos 4x) \, dx \\ &= \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c \\ &= \frac{1}{2} x + \frac{1}{8} \sin 4x + c \end{aligned} \quad (2)$$

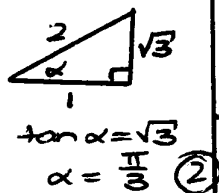
$$\begin{aligned} \text{b) } \tan(x+y) &= \frac{\tan x + \tan y}{1 - \tan x \tan y} \\ &= \frac{2 - \frac{1}{3}}{1 - (2)(-\frac{1}{3})} \\ &= \frac{6-1}{3+2} \\ &= 1 \\ x+y &= 45^\circ \end{aligned}$$

however,
 $0 < x < 90$ and $90 < y < 180$
 $\therefore 90 < x+y < 270$
 so $x+y$ is in 3rd quadrant
 as $\tan(x+y) > 0$

$$\therefore x+y = 225^\circ \quad (3)$$

$$\begin{aligned} \text{c) } f(x) &= x - \cos x \\ f'(x) &= 1 + \sin x \\ x_{n+1} &= x_n - \frac{x_n - \cos x_n}{1 + \sin x_n} \\ x_1 &= 0.5 - \frac{0.5 - \cos 0.5}{1 + \sin 0.5} \\ &= 0.7552224171 \\ &= 0.76 \text{ (to 2dp)} \end{aligned} \quad (3)$$

$$\begin{aligned} \text{d) (i) } \sin x + \sqrt{3} \cos x &= 2 \sin \left(x + \frac{\pi}{3} \right) \end{aligned}$$



$$\text{(ii) } \sin x + \sqrt{3} \cos x = 1$$

$$\begin{aligned} 2 \sin \left(x + \frac{\pi}{3} \right) &= 1 \\ \sin \left(x + \frac{\pi}{3} \right) &= \frac{1}{2} \\ x + \frac{\pi}{3} &= \frac{\pi}{6}, \frac{5\pi}{6} \\ x &= -\frac{\pi}{6}, \frac{\pi}{2} \\ \therefore x &= \frac{11\pi}{6}, \frac{\pi}{2} \end{aligned} \quad (2)$$

Question 4 (12)

$$\begin{aligned} \text{a) } \frac{1 + \cos 2\theta}{\sin 2\theta} &= \frac{1 + 2\cos^2 \theta - 1}{2\sin \theta \cos \theta} \\ &= \frac{2\cos^2 \theta}{2\sin \theta \cos \theta} \\ &= \frac{\cos \theta}{\sin \theta} \\ &= \cot \theta \end{aligned} \quad (2)$$

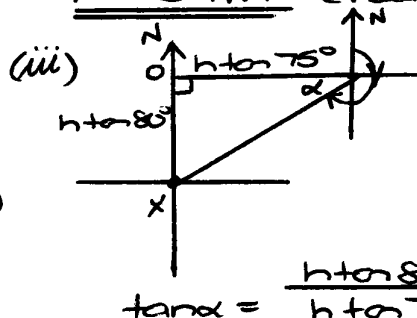
$$\begin{aligned} \text{b) } y &= \frac{1}{x} \\ \frac{dy}{dx} &= -\frac{1}{x^2} \cdot \frac{dx}{dx} \\ &= -\frac{2}{x^2} \\ \text{when } x &= 5, \frac{dy}{dx} = -\frac{2}{25} \end{aligned}$$

$\therefore y$ coordinate is decreasing
 at a rate of $\frac{2}{25}$ units/s (3)

$$\begin{aligned} \text{c) } \frac{OX}{h} &= \tan 80^\circ & \frac{OY}{h} &= \tan 75^\circ \\ OX &= h \tan 80^\circ & OY &= h \tan 75^\circ \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(ii) } OX^2 + OY^2 &= XY^2 \\ h^2 \tan^2 80^\circ + h^2 \tan^2 75^\circ &= 400^2 \\ h^2 &= \frac{400^2}{\tan^2 80^\circ + \tan^2 75^\circ} \\ h &= \sqrt{\frac{400^2}{\tan^2 80^\circ + \tan^2 75^\circ}} \end{aligned}$$

$$\begin{aligned} h &= 58.91812363 \\ h &= 59 \text{ m (nearest metre)} \end{aligned} \quad (3)$$



$$\begin{aligned} \alpha &= 56.65263461 \\ \alpha &= 57^\circ \text{ (nearest degree)} \end{aligned}$$

$$\therefore \text{Bearing} = 270 - 57 = 213^\circ \quad (2)$$

Question 5 (12)

a) $3x^3 - 17x^2 - 8x + 12 = 0$

$$\alpha\beta\gamma = -4$$

$$4\gamma = -4$$

$$\gamma = -1$$

$$(x+1)(3x^2 - 20x + 12) = 0$$

$$(x+1)(x-6)(3x-2) = 0$$

$$\underline{x = -1, 6, \frac{2}{3}} \quad (3)$$

b) $\frac{d}{dx}(\tan^3 x)$
 $= 3\tan^2 x \sec^2 x$
 $= 3(\sec^2 x - 1)\sec^2 x$
 $= \underline{3(\sec^4 x - \sec^2 x)} \quad (2)$

(iii) $\tan^3 x = 3 \int (\sec^4 x - \sec^2 x) dx$
 $3 \int \sec^4 x dx = \tan^3 x + 3 \int \sec^2 x dx$
 $= \tan^3 x + 3 \tan x + c$

$$\therefore \underline{\int \sec^4 x dx = \frac{1}{3} \tan^3 x + \tan x + c} \quad (2)$$

c) $T = 20 + Ae^{-kt}$
 $\frac{dT}{dt} = -kAe^{-kt}$
 $= -k(20 + Ae^{-kt} - 20)$
 $= \underline{-k(T - 20)} \quad (2)$

(ii) when $t=0, T=90$
 $\therefore 90 = 20 + A$
 $A = 70$

$$T = 20 + 70e^{-kt}$$

when $t=10, T=70$

$$70 = 20 + 70e^{-10k}$$

$$50 = 70e^{-10k}$$

$$e^{-10k} = \frac{5}{7}$$

$$-10k = \log \frac{5}{7}$$

$$k = -\frac{1}{10} \log \frac{5}{7}$$

$$(\quad = \frac{1}{10} \log \frac{7}{5})$$

when $t=15,$

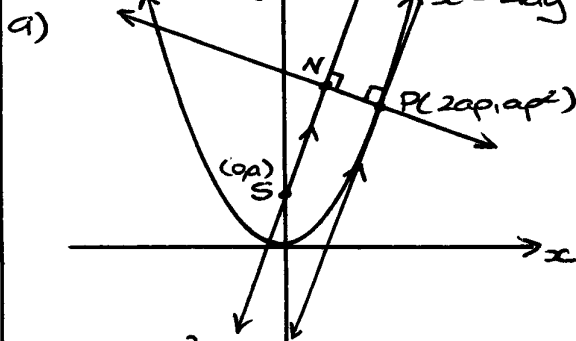
$$T = 20 + 70e^{-15k}$$

$$= 20 + 70\left(\frac{5}{7}\right)^{\frac{15}{10}}$$

$$= 62.25771274 \quad (3)$$

$$= \underline{62^\circ} \text{ (nearest degree)}$$

Question 6 (12)



(i) $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{x}{2a}$$

at P, $\frac{dy}{dx} = \frac{2ap}{2a} = p$

$$y - ap^2 = \frac{1}{p}(x - 2ap)$$

$$py - ap^3 = x + 2ap$$

$$\underline{x + py = ap^3 + 2ap} \quad (2)$$

(ii) NS: $m_{NS} = p$
 $y = px + a$

$$x + p^2x + ap = ap^3 + 2ap$$

$$(p^2 + 1)x = ap^3 + ap$$

$$= ap(p^2 + 1)$$

$$x = ap$$

$$\therefore y = ap^2 + a \quad (3)$$

N has coordinates $(ap, ap^2 + a)$

(iii) $x = ap$

$$p = \frac{x}{a}$$

$$y = ap^2 + a$$

$$= a\left(\frac{x}{a}\right)^2 + a$$

$$\underline{y = \frac{x^2}{a} + a} \quad (1)$$

b) Initial loan borrowed for n months =
 $= 30000(1.0075)^n$

1st repayment invested for $(n-1)$ months
 $= M(1.0075)^{n-1}$

2nd repayment invested for $(n-2)$ months
 $= M(1.0075)^{n-2}$

...

last repayment invested for 0 months
 $= M$

$$A_n = (\text{principal} + \text{interest}) - (\text{repayments} + \text{interest})$$

$$A_n = 30000(1.0075)^n - M(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$$

$$a = 1, r = 1.0075, n = n$$

$$A_n = 30000(1.0075)^n - M \left[\frac{1.0075^n - 1}{0.0075} \right] \quad (2)$$

(ii) $A_{60} = 0$

$$0 = 30000(1.0075)^{60} - M \left[\frac{1.0075^{60} - 1}{0.0075} \right]$$

$$M = \frac{30000(1.0075)^{60}(0.0075)}{1.0075^{60} - 1}$$

$$= 622.7506568$$

$$= \underline{\$622.75} \quad (1)$$

(iii) $A_{12} = 30000(1.0075)^{12} - 622.75 \left[\frac{1.0075^{12} - 1}{0.0075} \right] + 6000$

$$= 19025.10753$$

$$= \$19025.11$$

So

$$A_n = 19025.11(1.0075)^n - 622.75 \left[\frac{1.0075^n - 1}{0.0075} \right]$$

But $A_n = 0$

$$0 = 19025.11(1.0075)^n - 83033.3(1.0075^n - 1)$$

$$64008.223(1.0075)^n = 83033.3$$

$$(1.0075)^n = 1.297229153 \dots$$

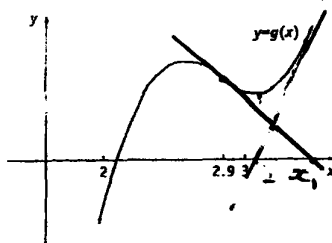
$$n = \frac{\log 1.297229153 \dots}{\log 1.0075}$$

$$= 34.82736244$$

$\therefore 35$ more repayments are needed (3)

Question 7 (12)

a)



As 2.9 is on the opposite side of the stationary point to the root, the tangent will cut the x-axis further from the root than the original approximation. (2)

b) Prove true for $n=1$

$$\text{LHS} = \frac{1}{1 \times 3} = \frac{1}{3}$$

$$\text{RHS} = \frac{1(3+5)}{4(2 \times 3)} = \frac{1}{3}$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence the result is true for $n=1$

Assume the result is true for $n=k$, where k is a positive integer

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{k(k+2)} = \frac{k(3k+5)}{4(k+1)(k+2)}$$

Prove the result is true for $n=k+1$

i.e. Prove

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(k+1)(k+3)} = \frac{(k+1)(3k+8)}{4(k+2)(k+3)}$$

Proof:

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{k(k+2)} + \frac{1}{(k+1)(k+3)}$$

$$= \frac{k(3k+5)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+3)}$$

$$= \frac{k(3k+5)(k+3) + 4(k+2)}{4(k+1)(k+2)(k+3)}$$

$$= \frac{3k^3 + 14k^2 + 19k + 8}{4(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)^2(3k+8)}{4(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)(3k+8)}{4(k+2)(k+3)}$$

Hence the result is true for $n=k+1$ if it is true for $n=k$

Since the result is true for $n=1$, then the result is true for all positive integral values of n by induction. (3)

$$(ii) \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k(k+2)}$$

$$= \lim_{n \rightarrow \infty} \frac{n(3n+5)}{4(n+1)(n+2)}$$

$$= \underline{\underline{\frac{3}{4}}}$$

①

$$c) (i) \frac{x+s}{h} = \frac{s}{D}$$

(ratio of sides in $\parallel \Delta$'s)

$$x+s = \frac{sh}{D}$$

$$x = \frac{sh}{D} - s$$

$$\underline{\underline{x = s \left(\frac{h}{D} - 1 \right)}} \quad (2)$$

$$(ii) \frac{dx}{dt} = \frac{ds}{dt} \left(\frac{h}{D} - 1 \right)$$

$$v = \frac{ds}{dt} \left(\frac{h}{D} - 1 \right)$$

$$= \frac{ds}{dt} \left(\frac{h-D}{D} \right)$$

$$\underline{\underline{\frac{ds}{dt} = \frac{Dv}{h-D}}}$$

②

$$(iii) \frac{ds}{dt} = \frac{1.2V}{5-1.2}$$

$$= \frac{1.2V}{3.8}$$

$$= \frac{6}{19} V$$

$$\therefore \underline{\underline{\frac{ds}{dt} < v}}$$

①

(iv) The mistake is that

$\frac{ds}{dt}$ measures the change in the length of the shadow Not the velocity of the shadow.

①